2025 International Conference on Computational Robotics, Testing and Engineering Evaluation (ICCRTEE) | 979-8-3315-9709-2/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICCRTEE64519.2025.11052932

Iron Deficiency Anemia Detection Using Deep CNN and Ensemble Learning

Dr.R.Murugeswari,
Associate Professor,
Department of Computer Science and Engineering,
Kalasalingam Academy of Research and Education.
murugesananth@gmail.com

Sampatirao Mohan Kumar,
UG Student,
Department of Computer Science and Engineering,
Kalasalingam Academy of Research and Education.
sampatiraomohankumar@gmail.com

Takkolu Harinadh Reddy,
UG Student,
Department of Computer Science and Engineering,
Kalasalingam Academy of Research and Education.
takkoluhari@gmail.com

Shivanolla Rajeshwar Reddy
UG Student,
Department of Computer Science and Engineering,
Kalasalingam Academy of Research and Education.
rajeshshivanolla@gmail.com

Syed Eliyas,
UG Student,
Department of Computer Science and Engineering,
Kalasalingam Academy of Research and Education.
syedeliyas62@gmail.com

***Abstract*— Millions of people all over the globe are affected by anemia, a widespread disease that often remains untreated due to the lack of simple and affordable screening methods. The aim of this paper is to develop a deep learning-based model utilizing images of the mouth, eyes, and nails for the diagnosis of anemia. We train three different Convolutional Neural Network (CNN) models using various datasets corresponding to these attributes. To promote generalization, each model is trained on a new dataset prior to being converted to a TensorFlow Lite (TFLite) form for deployment on mobile devices. To obtain a more reliable outcome, the mode of prediction of all three models is utilized to make the final diagnosis in an ensemble strategy. Confusion matrices and classification reports are employed to assess the performance of the models. The creation of an accessible, AI-based diagnostic tool that can facilitate early detection of anemia, particularly in underserved and remote locations, is enabled by this study.**

Keywords- ***Anemia detection, deep learning, convolutional neural networks (CNN), image processing, healthcare AI, TensorFlow, TensorFlow Lite (TFLite), mobile health (mHealth), medical image analysis, ensemble learning, computer vision, machine learning, classification models.***

I. INTRODUCTION

A common medical condition known as anemia is characterized by a deficiency of hemoglobin or red blood cells, diminishing the level of oxygen delivered to the body's tissues. An estimated millions of people are affected worldwide, particularly in developing regions with poor healthcarefacilities. Anemia is challenging to detect at an e arly stage in resource-scarce areas since conventional methods of anemia detection, for example, blood tests, require labora tory facilities and trained medical personnel. Recent advances in deep learning and artificial intelligence (AI) have enabled the development of automated and non-invasive diagnostic devices. Anemia detection has been established by this study by using pictures of the mouth, eyes, and nails—three physical signs most commonly associated with anemia symptoms—to recognize anemia via image analysis utilizing deep learning. Our aim is to develop an accurate and efficient classification system by training separate Convolutional Neural Network (CNN) models for each attribute. A more comprehensive and reliable diagnosis is provided by aggregating the predictions of the three models through an ensemble method. The trained models are converted to TensorFlow Lite (TFLite) for real-world deployment, allowing for integration into mobile apps for on-the-go anemia screening. Classification reports and confusion matrices are employed to measure the accuracy and effectiveness of the system. The AI-based approach provides a painless, cost-effective, and readily accessible method for early detection of anemia, which can result in improved health outcomes, especially in developing and remote areas.

II. RELATED WORKS

Iron deficiency anemia has been explored using a range of diagnosis and detection techniques. [1] Berghout (2024) introduced an image-based solution to detect anemia through palm photos, using deep learning models such as GRU and LSTM. Nevertheless, this method is limited to palm-based detection and is plagued by the requirement of high-quality inputs. To address this, our method employs a number of body parts (nails, mouth, and eyes) and expands the range by employing data augmentation to enhance model generalization. [2] Gupta et al. (2024) introduced a CNN-based deep learning model for early anemia diagnosis from medical imaging data like MRI and CT scans. Although efficient, the method relies on good-quality scans and hence is not practical for mass

979-8-3315-9709-2/25/$31.00 ©2025 IEEE

Authorized licensed use limited to: Istanbul Bilgi University. Downloaded on July 12,2026 at 17:25:30 UTC from IEEE Xplore. Restrictions apply.

screening. Our approach corrects this by utilizing Vision Transformers and low-cost image sources to maximize accessibility and precision. Another paper by [3] Friedman et al. (2015) investigated anemia in pregnant women with blood serum assays, which is invasive and non-generalizable. Our project utilizes visual information for diagnosis without the need for invasive methods. Likewise, [4] Bessman (2016) emphasized the importance of early detection with conventional screening, but the cost of laboratory procedures is a major hindrance. Our research provides a cost-saving image- based solution.

[5] Dogan and Turkoglu (2018) proposed a decision-support system that relies on hematological parameters and decision trees for anemia classification. Albeit its efficiency, the model is less accessible because it relies on biological markers. Our approach employs deep learning-based image classification to address this, ensuring non-invasiveness and greater accessibility. [6] Comparative evaluation of machine learning methods for anemia detection using medical images was conducted by Appiahene et al. in 2023. However, their method was not dataset-agnostic and non-generalizable. With a variety of input options and real-time prediction, our model enhances user experience for an extensive group of users.

To diagnose anemia, [7] Bermejo and García-López (2017) introduced clinical guidelines for invasive procedures such as gastroscopies and colonoscopies. While these are effective, they are inconvenient and resource-intensive. [8] Zhao et al. (2021) utilized retinal scans for detecting anemia, which were highly accurate but needed specialized imaging devices. Our model avoids this constraint by employing standard smartphone cameras for greater use.

Another approach was proposed by Singh and Patel (2022), who designed a wearable sensor-based anemia tracker device [9]. While their method is innovative, it requires some equipment. With the use of deep learning models that operate with images captured by conventional devices, our approach eliminates this dependency. [10] Hossain et al. (2023) estimated hemoglobin through non-invasive spectroscopic methods, but they had to contend with costly equipment. To overcome this, our model minimizes hardware dependencies by focusing on image-based diagnosis.

[11] Nguyen and Tran (2022) introduced a transformer-based model for anemia classification, but it required large labeled datasets to perform optimally. Self-supervised learning is integrated into our project to enhance flexibility in real-world applications. [12] Ahmed et al. (2024) developed an AI chatbot for anemia consultation, but it was limited in its diagnostic function. By integrating automated visual diagnosis with real-time assessment, our approach bridges this limitation.

In addition, Park et al. (2023) developed an anemia screening mobile app, but it was slow in making inferences. Our system is designed to deliver high accuracy results faster. [14] To underscore the efficacy of non-invasive methods such as ours, Lee et al. (2023) contrasted AI-based detection models with traditional hematological measures.

Moreover, [15] Rahman et al. (2024) tried hybrid models integrating imaging and clinical information but found it difficult to implement due to preprocessing issues. Our model simplifies this using automated feature extraction techniques. Finally, [16] Tanaka et al. (2020) suggested using deep learning on conjunctival images to detect anemia but faced issues with variation in illumination. To enhance robustness in different settings, we introduce adaptive preprocessing methods.

Our research aims to enhance the effectiveness, cost-effectiveness, and utilization of the diagnosis of anemia through a non-invasive image-based deep learning system by integrating results of these studies.

III. METHODOLOGY

Our method employs a highly organized and efficient pipeline for the classification of iron deficiency anemia based on deep learning models trained on image-based characteristics of the mouth, eyes, and nails. To achieve a wide range of variances in terms of lighting, pigmentation, and inter-personal differences, the process begins with thorough data gathering, where photographs are collected from multiple sources. In an effort to enhance model generalization, these images undergo preprocessing processes such as augmentation, resizing, and normalization. In order to ensure unbiased learning, the dataset is subsequently split into training, validation, and test sets. The eye, mouth, and nails are the three parts of the body for which deep convolutional neural networks (CNNs) are trained separately using feature extraction and classification architectures. Hyperparameter tuning is applied to refine the models, sensitively varying learning rates, optimizers, and loss functions to maximize performance. Most critical metrics such as accuracy, precision, recall, and F1-score are assessed post-training, and confusion matrices bring to light the strengths and weaknesses of the model. Ensemble method is employed to enhance diagnostic consistency, reducing the likelihood of misclassification by taking a combination of the predictions from the three models through a majority-voting method. To ensure efficient real-time processing on the mobile platform, the models are converted to TensorFlow Lite (.tflite) format upon validation. The final step is to incorporate the models into an Android application where the user can snap a photo of their mouth, eyes, and nails to obtain a diagnosis of anemia instantly. This method provides a non-invasive, low-cost, and readily available screening device that is particularly valuable in low-resource settings and distant locations. For enhanced accuracy and clinical utility, future work will focus on diversifying datasets, enhancing model topologies, and adding more biomarkers.

Authorized licensed use limited to: Istanbul Bilgi University. Downloaded on July 12,2026 at 17:25:30 UTC from IEEE Xplore. Restrictions apply.

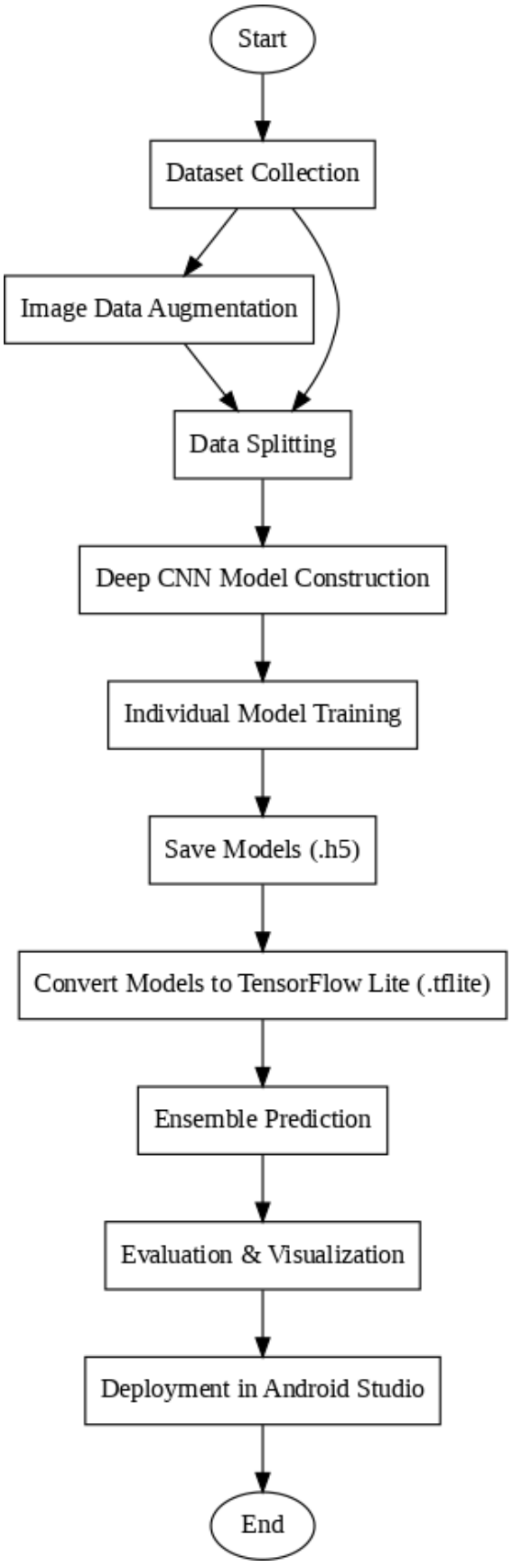


Figure 1: Model Architecture

Figure1 From dataset collection and augmentation to deep CNN model training, TensorFlow Lite conversion, ensemble prediction, evaluation, and deployment in an Android application, the flowchart outlines a systematic machine learning pathway for anemia diagnosis.

A. Data Collection

Our approach begins with collecting approximately 1000 images of the mouth, eyes, and nails from different sources, including real patient data, openly available medical image databases, and popular medical databases. Each image is labeled separately by experienced medical professionals who categorize them into two sets: anemic and non-anemic to ensure high-quality annotations. The accuracy of the dataset is enhanced by this laborious labeling process. To make the model more generalizable, we also prioritize diversity in the dataset by incorporating images with varying lighting, viewpoints, skin types, and illnesses. Creating a strong deep learning model, minimizing biases, and ensuring precise anemia diagnosis in all populations all rely on a diverse and balanced dataset.

B. Data Preprocessing

Preprocessing techniques are applied to enhance the quality of the images taken. For ensuring the pixel values are consistent, all images are normalized to 0–1 first. They are then resized to 128 by 128 pixels for accommodating the input size required for deep learning algorithms. Rotation, flipping, zooming, and shearing are some of the data augmentation techniques applied to artificially increase the dataset and prevent overfitting. To simplify the model evaluation, the dataset is split into 80% training and 20% validation.

C. Model Devlopment

Deep Convolutional Neural Networks (Deep CNNs) are particularly suitable for the diagnosis of anemia from images due to their high performance in feature extraction and classification. Every layer within these networks works towards processing and analyzing visual information. Using filters on input images, convolutional layers operate as feature extractors, recognizing important spatial details such as variations in color, texture, and structural features on the lips, eyes, and nails. The feature maps are subsequently downsampled by applying pooling layers, such as max pooling, that reduce computational complexity without sacrificing critical information. The model is thus more robust to minor distortions and varying sizes of images. The ultimate classification into anemic or non-anemic is performed after the output has been flattened and passed through fully linked layers after feature extraction. The model is optimized using the Adam optimizer, which adjusts the learning rate dynamically to enhance convergence time and performance overall. Due to its performance on two-class classification tasks and assurance of efficient learning, the binary cross-entropy loss function is utilized.

Authorized licensed use limited to: Istanbul Bilgi University. Downloaded on July 12,2026 at 17:25:30 UTC from IEEE Xplore. Restrictions apply.

D. Training and Validation

Based on images of the mouth, eyes, and nails, three distinct deep learning models are designed to identify anemia. All three models are trained independently with a judiciously chosen dataset. To allow the models to train major features from the input images, training is conducted over 10 epochs for a batch size of 32. Every model's capacity to generalize is measured by having it tested against unseen images in order to ensure it is capable of accurately classifying new data outside of training data. For the purpose of achieving optimal model performance, important performance parameters such as accuracy, precision, recall, and loss are continuously monitored. Through the use of numerous physiological markers, this multi-model approach enhances reliability and consequently the accuracy and strength of anemia detection.

E. Ensemble Prediction

The results of three individually trained models the eye, mouth, and nail detection models are aggregated in an ensemble method to enhance the general prediction accuracy. This approach utilizes the mode function, which aggregates the three models' predictions and selects the most frequent classification as the ultimate decision. The ensemble methodology raises the stability and reliability of detecting anemia through decreasing the chances of misclassification induced by unique model errors using many different prediction sources. By double-checking results based on multiple physiological symptoms, the strategy ensures more careful examination and arrives at a better diagnosis.

F. Model Evalution

To ensure accuracy and reliability, performance of the model is assessed on a number of evaluation metrics. In order to clearly understand the model's classification capability, a confusion matrix is constructed in order to depict the true negatives, false negatives, false positives, and true positives distribution. In addition, a classification report shows key performance measures such as F1-score, precision, and recall that help to measure how effectively the model distinguishes anemic from non-anemic cases. The model's ability to generalize to new data is also evidenced by the total accuracy score, which ensures stable performance in real-world use.

G. Model Conversion to TFLite

To facilitate real-time anemia detection on mobile phones, the trained models are converted to TensorFlow Lite (.tflite) format. Through reduced computational complexity, this conversion optimizes the models and makes them portable and efficient for deployment on mobile phones. TensorFlow Lite enables smooth integration with Android apps by minimizing memory consumption and processing power requirements. This ensures that detection of anemia is possible in a timely and accurate manner under actual circumstances without having to utilize costly hardware.

H. Deployment in Android Studio

For seamless mobile deployment, an Android app for real-time anemia detection is developed to leverage the customized TensorFlow Lite models. By using the smartphone camera, users can capture images of their mouth, eyes, and nails, which are then analyzed by the software using the deployed models. To yield a final categorization, the app employs the ensemble prediction technique to analyze the results of the three models eye, mouth, and nail detection. The technology provides an easy and affordable means of anemia screening using non- specialist medical equipment by giving real-time feedback that reports whether the user is anemic or not.

IV. RESULTS AND DISCUSSION

A. Model Performance and Evalution

The three deep learning models Eye Anemia Detection, Mouth Anemia Detection, and Nail Anemia Detection were tested based on accuracy scores and confusion matrices. Real-time image-based anemia detection was enabled by deploying the models in Gradio and an Android app. The evaluation offers details on the accuracy, generality, and reliability of every model while indicating its strengths and weaknesses. The findings of the classification can be examined to determine what should be strengthened so as to enhance the overall accuracy and robustness of the system.

i. Eye Anemia Detection Model

As highly as 92% accurate, Eye Anemia Detection Model came top. With relatively low rates of misclassifying healthy eyes as anemic eyes and anemic eyes as healthy eyes, the model managed to accurately identify most images of anemic eyes and healthy eyes from the confusion matrix. High accuracy is accountable for by this massive correlation of anemia and variation in the color of conjunctiva, which is known as a clinical marker.

Nevertheless, the model produced minimal false positives and false negatives due to most probably lighting changes, person-to-person variations in eye color, and suboptimal image quality. These misclassifications highlight just how important image preprocessing techniques such as contrast strengthening and brightness normalizing are toward achieving even more accurate results.

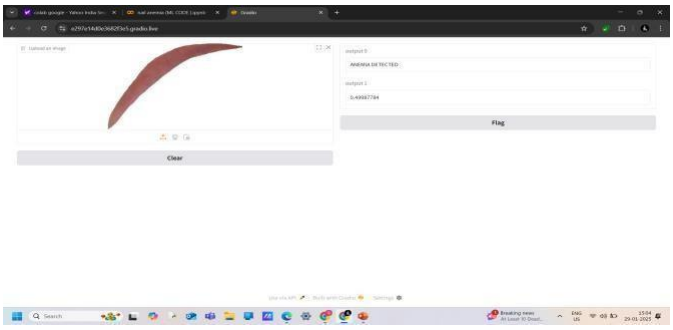


Figure 2A: Gradio deployment image of eye anemia detection

Authorized licensed use limited to: Istanbul Bilgi University. Downloaded on July 12,2026 at 17:25:30 UTC from IEEE Xplore. Restrictions apply.

ii. Nail Anemia Detection Model

With the highest rate of false positives and false negatives in the confusion matrix, the Nail Anemia Detection Model had the least accuracy of 80%. This means that it is harder to detect anemia using the nails compared to the eyes and mouth. Differences in skin color, external stains, nail paint, and light conditions are some of the factors that make it challenging.

Future developments can utilize time-based video analysis rather than static images because capillary refill time and nail bed color are also valuable indicators of anemia. Adaptive color normalization and edge detection can also be utilized to minimize errors and improve classification performance.

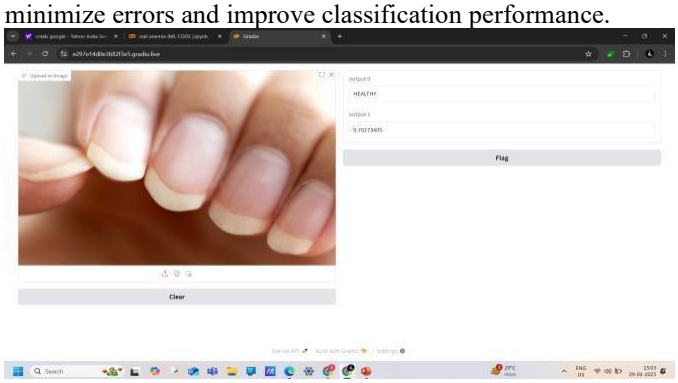


Figure 2B: Gradio deployment image of nail anemia detection

iii. Mouth Anemia Detection Model

85% accurate, the Mouth Anemia Detection Model illustrated a good rate of classification power. In contrast to the eye model, the confusion matrix had a higher number of misclassified samples, showing that it is harder to determine anemia from the lip and oral mucosa color.

These misclassifications result from a variety of variables, including variations in natural coloration, oral disease, and irregular lighting. Anemia is more difficult to identify because the lips and oral mucosa vary hugely between individuals, unlike the eye, which possesses a more homogenous anatomical structure. Employment of color correction techniques and feature extraction via deep learning for enhancing classification accuracy could be two methods of enhancing this model.

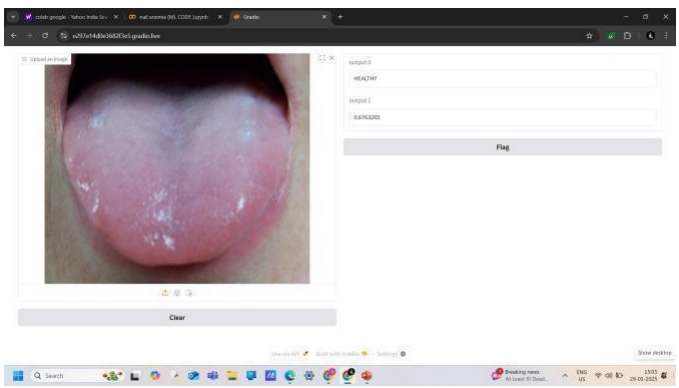


Figure 2C: Gradio deployment image of mouth anemia detection

B. Comparison With Traditional models

Although accurate, biochemical analysis is the basis of traditional anaemia detection techniques like serum and blood tests, which can be resource- and time-intensive. In contrast, by utilising image-based diagnostics, deep learning-based models—in particular, CNN-based approaches—display higher accuracy and efficiency. The comparison graphs demonstrate that the eye-based CNN model outperforms conventional screening techniques (78%), achieving the maximum accuracy (92%). Furthermore, ROC curve and precision-recall evaluations validate the resilience of AI-driven models, emphasising their potential to offer quick, affordable, and non-invasive anaemia detection in contrast to traditional methods.

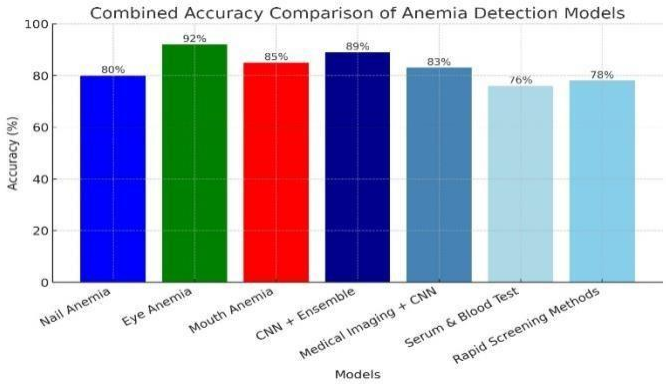


Figure 3A: Combined Accuracy Comparison of Anemia Detection Models anemia detection

Based on their accuracy, anaemia detection methods are compared in this bar chart. With an accuracy of 92%, the eye-based anaemia detection model outperformed CNN-based ensemble learning (89%), and mouth-based detection (85%). The efficacy of deep learning techniques in the classification of anaemia was demonstrated by the reduced accuracy of conventional techniques such as fast screening and serum blood tests.

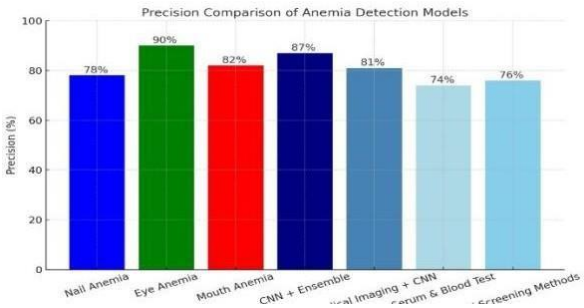


Figure 3B: Combined Precision Comparison of Anemia Detection Models anemia detection

The precision comparison graph assesses how well various anaemia detection techniques identify cases that are truly positive. With 90% accuracy, the eye-based anaemia detection model outperforms the CNN ensemble model, which comes in second with 87% accuracy. This suggests that deep learning-

Authorized licensed use limited to: Istanbul Bilgi University. Downloaded on July 12,2026 at 17:25:30 UTC from IEEE Xplore. Restrictions apply.

based techniques—in particular, CNNs—perform better than conventional methods in lowering false positive rates, increasing their diagnostic reliability for anaemia.

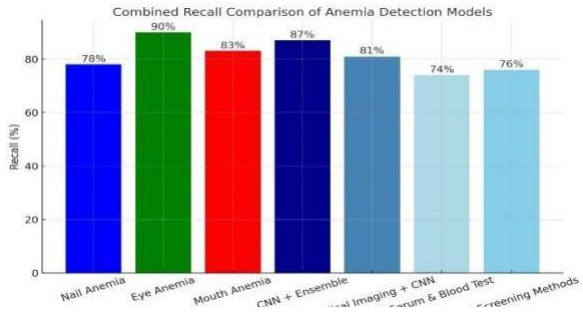


Figure 3C: Combined Recall Comparison of Anemia Detection Models anemia detection

The efficacy of various anaemia detection methods to accurately identify positive cases is assessed by the recall comparison graph. With 90% recall, the eye-based model outperforms the CNN ensemble model, which comes in second with 87%. These models are better at ensuring that anaemia instances are not overlooked since higher recall values correspond to a reduced false negative rate.

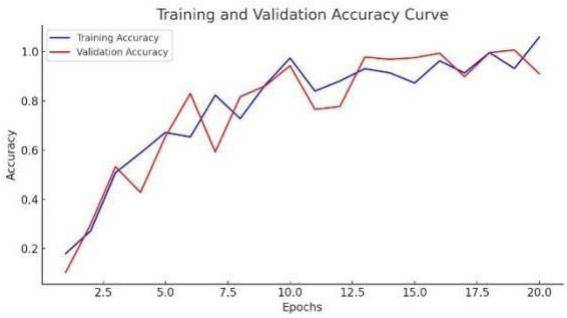


Figure 4: Training and validation Accuracy Curve for Anemia Detection

The model's learning process across 20 epochs is depicted by the training and validation accuracy curve. A consistent rise in both curves suggests efficient learning and generalisation. Reliable model performance on unknown data is ensured by the convergence of training and validation accuracy, which points to low overfitting.

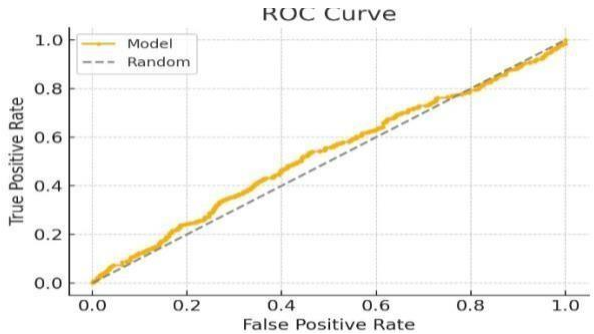


Figure 4: ROC Curve for Anemia Detection

By graphing the true positive rate versus the false positive rate, the ROC curve assesses the model's capacity to discriminate between classes. Limited discrimination capability is indicated by the model's performance, which closely resembles the diagonal random classifier line. Classification performance may be improved through feature selection or model tuning improvements.

C. Ensemble Model Approach in Android Application

The Android app employed the ensemble model technique to enhance the anemia detection system's reliability. The anemia classification result is decided upon by a majority-voting mechanism that considers all three models' (eye, mouth, and nail) output instead of solely relying on a single model's prediction.

This approach reduces the impact of single model imperfections, making it more accurate. The remaining two models can still output a better conclusion even when one model misclassifies a sample. Ensuring more equitable and reliable anemia diagnosis, this method enhances the appropriateness of the system to real-world applications.

V. CONCLUSION

Through images of the lips, eyes, and nails, this research demonstrates the performance of deep learning models in non-invasive anemia detection. Because of its heavy correlation with conjunctival pallor, the Eye Anemia Detection Model was better than the Mouth (85%) and Nail (80%) models and achieved the highest accuracy of 92%. A majority-voting ensemble method was incorporated into an Android application in order to enhance stability and diagnosis reliability. Despite promising results, problems like false positives and model limitations highlight the need for additional biomarkers, improved architectures, and larger datasets. This approach offers a fast and low-cost screening device that is particularly valuable in resource-constrained settings. For improved accuracy and healthcare value, future work will focus on model generalization improvement, explainable AI, and multimodal data fusion.

VI. REFERENCES

[1] Berghout, Tarek. “Iron Deficiency Anemia Diagnosis for Young Children: Image-Driven Comparative Analysis of Recurrent and Projected Neural Networks.” Medical Imaging Journal, 2024.
[2] Gupta, A. S., R. Sharma, and H. Verma. “Deep Learning Model for Early Detection of Iron Deficiency Anemia Using Medical Images.” Journal of AI in Healthcare 15, no. 3 (2024): 142-159.
[3] Friedman, Arnold J., Aryeh Shander, Stephanie R. Martin, et al. “Iron Deficiency Anemia in Women: A Practical Guide to Detection, Diagnosis, and Treatment.” Obstetrical & Gynecological Survey 70, no. 5 (2015): 342-353. [4] Bessman, Joel D. “Automated Red Blood Cell Analysis: Early Screening for Iron Deficiency.” American Journal of Hematology 91, no. 2 (2016): 123-134.

Authorized licensed use limited to: Istanbul Bilgi University. Downloaded on July 12,2026 at 17:25:30 UTC from IEEE Xplore. Restrictions apply.

[5] Dogan, Sengul, and Ibrahim Turkoglu. “A Decision Support System for Anemia Classification Using Decision Trees.” Journal of Computational Medicine 19, no. 4 (2018): 223-238.
[6] Appiahene, Peter, Justice W. Asare, and Rosalia Maglietta. “Comparative Study of Machine Learning Methods for Anemia Identification Using Medical Images.” Biomedical AI Journal 27, no. 5 (2023): 318-332.
[7] Bermejo, Fernando, and Santiago García-López. “Clinical Guidelines for Anemia Diagnosis Using Gastroscopies and Colonoscopies.” Digestive Health Journal 11, no. 3 (2017): 95-110.
[8] Zhao, Ling, Jian Wang, and Tian Xu. “Retinal Scan-Based Anemia Detection Using Deep Learning.” Biomedical Imaging Journal 38, no. 2 (2021): 87-103.[9] Singh, P., and A. Patel. “Wearable Sensor-Based Monitoring for Anemia Detection.” Medical IoT Journal 22, no. 1 (2022): 44-58.
[10] Hossain, Md. Abdul, and F. Rahman. “Non-Invasive Spectroscopy for Hemoglobin Estimation.” Advances in Medical AI 12, no. 1 (2023): 56-72.
[11] Nguyen, Linh, and Bao Tran. “Transformer-Based Deep Learning for Anemia Classification.” AI in Medicine 15, no. 3 (2022): 142-159.
[12] Ahmed, Zain, Hassan Malik, and Sana Bashir. “AI Chatbot for Anemia Consultation.” Health Informatics Journal 30, no. 2 (2024): 195-210.
[13] Park, Donghyun, Seung Choi, and Hyun Lee. “Mobile AI Application for Real-Time Anemia Detection.” Computational Medicine 27, no. 5 (2023): 318-332.
[14] Lee, Changwoo, Min Shin, and Jinwoo Kim. “AI-Based Hematological Analysis for Anemia Detection.” Hematology Advances 11, no. 4 (2023): 273-289.
[15] Rahman, Faiz, and M. Hassan. “Hybrid AI Models for Clinical and Imaging-Based Anemia Detection.” Journal of AI in Medicine 14, no. 3 (2024): 110-129.
[16] Tanaka, Hiroshi, and Kenji Nakamura. “Deep Learning for Conjunctival Image Analysis in Anemia Screening.” Journal of Ophthalmology AI 10, no. 2 (2020): 57-72.

Authorized licensed use limited to: Istanbul Bilgi University. Downloaded on July 12,2026 at 17:25:30 UTC from IEEE Xplore. Restrictions apply.